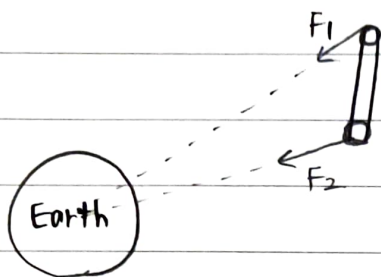
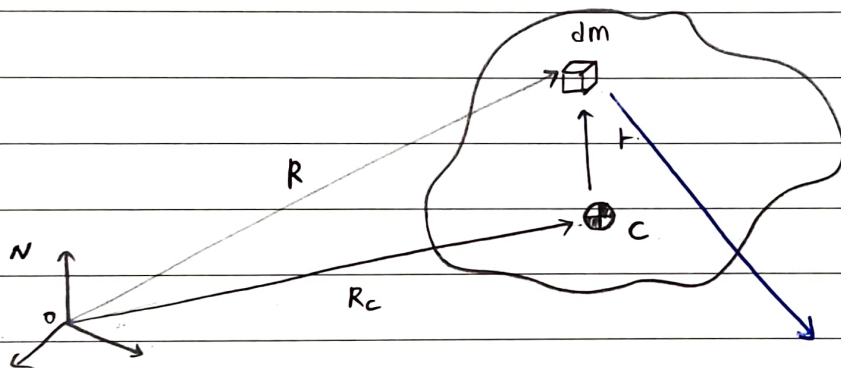


< Gravity Gradient Torque Development 6/29 >



작용하는 중력의 크기가 다름.

* Gravity gradient Torque.



$$dF_g = - \frac{G M_e dm}{|R|^2} \frac{R}{|R|}$$

1) Gravity gradient torque acting on the spacecraft

$$L_G = \int_B r \times dF_g$$

$$= - \int_B r \times \frac{G M_e}{|R|^3} (R_C + r) dm$$

$$= G M_e R_C \times \int_B \frac{r}{|R|^3} dm$$

$|R|^{-3} = |R_C + r|^{-3}$ 에서 r 은 R_C 에 비해 매우 작음. 따라서

테일러 급수로 변형 후. 1st order만 두고 나머지는 0으로 근사.

* $|R|^{-3}$ 근사

$$f(r) = R^{-3}$$

$$f(x) = f(x_0) + \frac{1}{1!} \frac{\partial f}{\partial x} \Big|_{x=x_0} \cdot \Delta x + \dots$$

r 에 대해서 1st order term 만 고려해 보자면 $f(r) = f(r=0) + \frac{1}{1!} \frac{\partial f}{\partial r} \Big|_{r=0} \cdot r$

$$f(r=0) = R_c^{-3}$$

$$\frac{\partial f}{\partial r} = -3R^{-4} \frac{\partial R}{\partial r}$$

$$= -3R^{-5} (R_c + r)$$

$$\frac{\partial f}{\partial r} \Big|_{r=0} = -\frac{3}{R_c^5} \vec{R}_c$$

$$R^2 = R \cdot R = (R_c + r) \cdot (R_c + r)$$

$$2R \frac{\partial R}{\partial r} = \frac{\partial}{\partial r} (R_c \cdot R_c + 2r \cdot R_c + r \cdot r)$$

$$= 2R_c + 2r$$

$$\frac{\partial R}{\partial r} = \frac{1}{R} (R_c + r)$$

L_G 를 다시 적으면

$$L_G = G M_e \vec{R}_c \times \int_B \frac{r}{|R|^3} dm$$

$$= G M_e \vec{R}_c \times \int_B r \left(R_c^{-3} - \frac{3}{R_c^5} \vec{R}_c \cdot r \right) dm$$

$$= \frac{G M_e}{R_c^3} \vec{R}_c \times \int_B r \left(1 - \frac{3 \vec{R}_c \cdot r}{R_c^2} \right) dm$$

since $\int_B r dm = 0$

$$L_G = \frac{3 G M_e}{R_c^5} \vec{R}_c \times \int_B -\vec{r} (\vec{r} \cdot \vec{R}_c) dm$$

$$a \times (b \times c) = (a \cdot c)b - (a \cdot b)c \text{ 에서 변형하면}$$

$$-(a \cdot c)b = -a \times (b \times c) - (a \cdot b)c$$



$$-(r \cdot R_c)r = -r \times (r \times R_c) - (r \cdot r)R_c$$



$$L_G = \frac{3GM_e}{R_c^5} \vec{R}_c \times \int_B -(r \times (r \times R_c) + (r \cdot r)R_c) dm$$

$$= \frac{3GM_e}{R_c^5} \vec{R}_c \times \left(\int_B -[\ddot{r}][\ddot{r}] dm \right) \vec{R}_c - \frac{3GM_e}{R_c^5} \left(\int_B r^2 dm \right) \vec{R}_c \times \vec{R}_c$$

Using the matrix definition.

$${}^B[I_c] = \int_B -[\ddot{r}][\ddot{r}] dm = \int_B \begin{bmatrix} \ddot{r}_2^2 + \ddot{r}_3^2 & -\ddot{r}_1 \ddot{r}_2 & -\ddot{r}_1 \ddot{r}_3 \\ -\ddot{r}_1 \ddot{r}_2 & \ddot{r}_1^2 + \ddot{r}_3^2 & -\ddot{r}_2 \ddot{r}_3 \\ -\ddot{r}_1 \ddot{r}_3 & -\ddot{r}_2 \ddot{r}_3 & \ddot{r}_1^2 + \ddot{r}_2^2 \end{bmatrix} dm$$

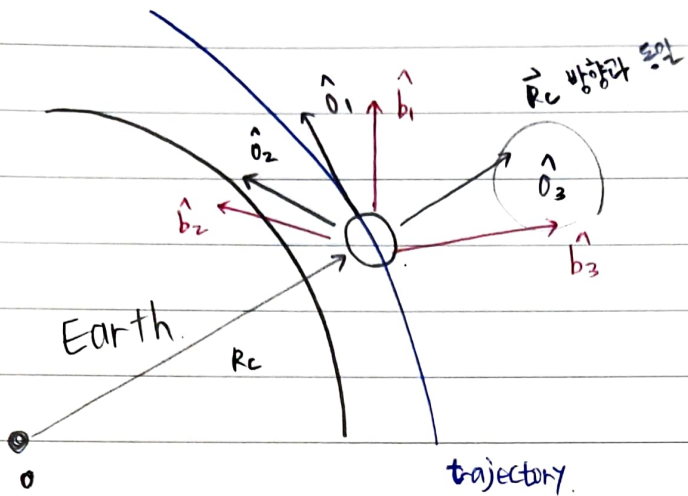
* the gravity torque on a rigid body (1차 근사 가정)

$$L_G = \frac{3GM_e}{R_c^5} \vec{R}_c \times [I] \vec{R}_c$$

[I]가 스칼라 x Identity 일때 $L_G = 0$, 이때 spacecraft는 sphere or cube.

좌표계에 상관없이 항상 성립하는 식.

< Gravity Gradient Torque in body frame b/29 >



$$O: \{ \hat{o}_1, \hat{o}_2, \hat{o}_3 \}$$

(orbit frame)

$$\vec{R}_c = R_c \hat{o}_3 = \begin{bmatrix} 0 \\ 0 \\ R_c \end{bmatrix}$$

$$R_c = \begin{bmatrix} R_{c1} \\ R_{c2} \\ R_{c3} \end{bmatrix} = [B0] \begin{bmatrix} 0 \\ 0 \\ R_c \end{bmatrix} \quad : [I] \text{ 은 회전하더라도 결국 body frame으로 옮겨서 봐야함}$$

$$L_G = \frac{3GM_e}{R_c^5} \begin{bmatrix} R_{c1} \\ R_{c2} \\ R_{c3} \end{bmatrix} \times \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} R_{c1} \\ R_{c2} \\ R_{c3} \end{bmatrix}$$

$$= \begin{bmatrix} L_{G1} \\ L_{G2} \\ L_{G3} \end{bmatrix} = \frac{3GM_e}{R_c^5} \begin{bmatrix} R_{c2}R_{c3}(I_{33}-I_{22}) \\ R_{c1}R_{c3}(I_{11}-I_{33}) \\ R_{c1}R_{c2}(I_{22}-I_{11}) \end{bmatrix}$$

* L_G 를 0으로 만드는 방법

$$① I_{11} = I_{22} = I_{33}$$

$$② \vec{R}_c = R_c \hat{b}_i \quad (\vec{R}_c \text{ 방향과 body의 } I_{11}, I_{22}, I_{33} \text{ 중 하나의 벡터의 방향이 동일})$$

$$\left(\begin{array}{l} i=3 \text{ 이라면 } \vec{R}_c \text{ 를 } R_c \hat{b}_3 \text{ 로 표현 가능해 } R_{c2}, R_{c1} \text{ 이 } 0, \text{ 따라서} \\ R_{c1}R_{c3}, R_{c2}R_{c1}, R_{c3}R_{c2} \text{ 모두 } 0 \text{ 이 되어 } L_G \text{도 } 0 \end{array} \right)$$

< Gravity Gradient: Net Spacecraft Force 6/29 >

' Altitude 변화가 Orbit에도 변화를 불러온다 '

$$F_G = \int_B dF_G = -G M_e \int_B \frac{R}{|R|^3} dm$$

Using the simplifying assumption

$$\frac{1}{|R|^3} \approx \frac{1}{R_c^3} \left[1 - \frac{3}{2} \left(2 \frac{R_c \cdot r}{R_c^2} + \frac{r \cdot r}{R_c^2} \right) + \frac{15}{2} \frac{(R_c \cdot r)^2}{R_c^4} \right]$$

torque에서는 1차항까지만 사용하지만
force에서는 1차항이 소거되어 2차항까지 보아야 함

$$F_G = -\frac{G M_e}{R_c^3} \left[\int_B r dm - \frac{3}{R_c^2} \int_B (r \cdot R_c) r dm - \frac{3}{R_c^2} \int_B (R_c \cdot r) R_c dm \right. \\ \left. + R_c \int_B dm - \frac{3}{2 R_c^2} \int_B R_c (r \cdot r) dm + \frac{15}{2 R_c^4} \int_B (R_c \cdot r)^2 R_c dm \right]$$

$$-(r \cdot R_c) r = -r \times (r \times R_c) - (r \cdot r) R_c \text{ 이므로}$$

$$F_G = -\frac{G M_e}{R_c^3} \left[m \vec{R}_c - \frac{3}{R_c^2} \int_B (\vec{r} \times (\vec{r} \times \vec{R}_c) + r^2 \vec{R}_c) dm \right. \\ \left. - \frac{3}{2 R_c^2} \int_B r^2 \vec{R}_c dm + \frac{15}{2 R_c^4} \int_B \vec{R}_c \cdot (\vec{r} \times (\vec{r} \times \vec{R}_c) + r^2 \vec{R}_c) \vec{R}_c dm \right]$$

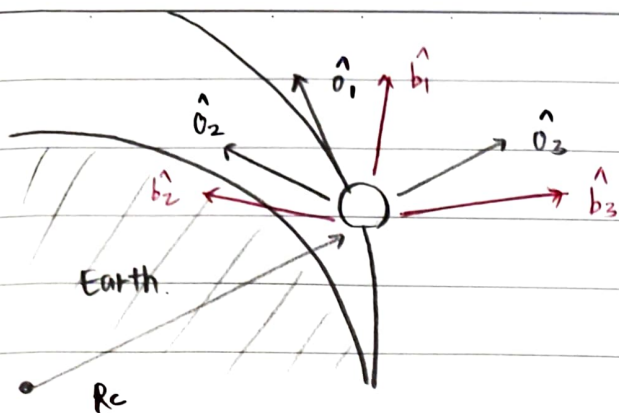
note the identities

$$\int_B r^2 dm = \frac{1}{2} \text{tr}([I]) \quad \hat{r} = \vec{R}_c / R_c$$

* the gravity force is finally written as

$$\vec{F}_G = - \frac{\mu m}{R_c^3} \left[1 + \frac{3}{m R_c^2} \left([I] + \frac{1}{2} \left(\text{tr}([I]) - 5 (\hat{r}^T [I] \hat{r}) \right) [I_{3 \times 3}] \right) \right] \vec{R}_c$$


< Gravity gradient relative equilibria orientations b/30 >



* equations of motion.

$$[I] \dot{W}_{B/N} + [\tilde{W}_{B/N}][I]W_{B/N} = L_c$$

$$W_{B/N} = W_{B/O} + W_{O/N}$$

$$W_{O/N} = n \hat{o}_2$$

$$W_{B/O} = 0$$

매순간 각도가 틀어지고 변하는 orbit frame에 대해
body frame의 상대적 attitude가 일정.
Relative equilibria condition

① $L_q = 0$ 이 되기 위한 조건

$$\hat{b}_3 = \hat{o}_3 : \text{Principal axes 중 하나가 } \hat{o}_3 \text{과 일치}$$



$$[I] = \begin{bmatrix} I_{11} & I_{12} & 0 \\ I_{12} & I_{22} & 0 \\ 0 & 0 & I_{33} \end{bmatrix}$$

: I_3 축에 대해서는 body의 $\hat{b}_3$ 와 $\hat{o}_3$ 가 일치하나

I_1, I_2 축에 대해서는 body와 orbit의 상대적 frame

차이가 발생하므로 orbit frame에서 볼 시, I_{12} 와 I_{21} 이 발생.

② Angular velocity condition. (relative equilibria가 되기 위한 조건)

$$\omega_{B/O} = 0 \quad \dot{\omega}_{B/O} = 0$$

$$\omega_{O/N} = n \hat{o}_2 \quad \dot{\omega}_{O/N} = 0$$

↓

$$[\tilde{\omega}_{B/N}][I]\omega_{B/N} = 0$$

↓

$$^0 \begin{pmatrix} 0 \\ 0 \\ -I_{12}n^2 \end{pmatrix} = 0$$

↓

$$I_{12} = 0$$

relative equilibria는 항상 principal axes가 orbit frame과 나란해야 한다.

이러야지만 principal axes에 대해 ω frame이
존재 $I_{12} = 0$ 이 결과로 나오기 때문.

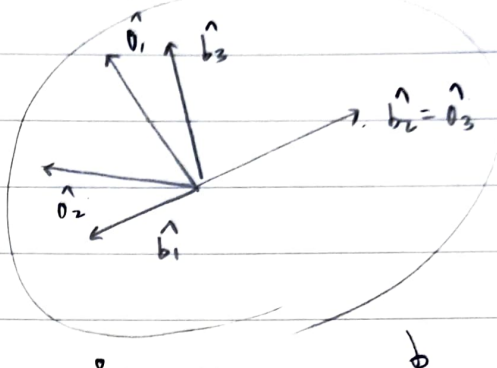
* 결과적으로 relative equilibria를 이루기 위해서는 $L_H = 0$, $[I]\omega_{B/N} = 0$,

$[\tilde{\omega}_{B/N}][I]\omega_{B/N} = 0$ 이어야 한다.

<concept check 2 - Gravity Gradient Equilibria 6/30>

1.

$${}^0 \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{bmatrix} \text{ where } \hat{b}_2 = \hat{b}_3$$



$$W_{B/O} = 0, \quad \dot{W}_{B/O} = 0.$$

$$W_{B/N} = \begin{bmatrix} 0 \\ n \\ 0 \end{bmatrix}$$

$\hat{o}_3$ is the direction of

the principal

axes and the direction of the

$$W_{O/N} = n \hat{o}_2, \quad \dot{W}_{O/N} = 0.$$

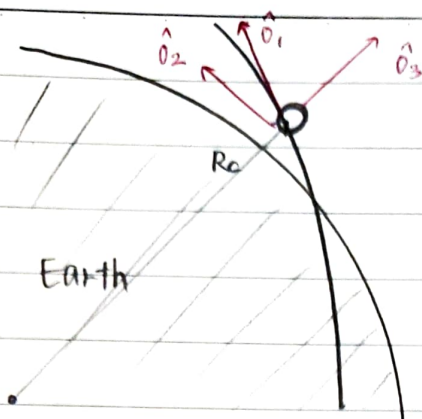
$$[\tilde{W}_{B/N}]^0 [I] W_{B/N}$$

$$= \begin{bmatrix} 0 & 0 & n \\ 0 & 0 & 0 \\ -n & 0 & 0 \end{bmatrix} \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{bmatrix} \begin{bmatrix} 0 \\ n \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & n \\ 0 & 0 & 0 \\ -n & 0 & 0 \end{bmatrix} \begin{bmatrix} n I_{12} \\ n I_{22} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ -n^2 I_{12} \end{bmatrix}$$

< Gravity gradient: linear stability about equilibria



Roll: about $\hat{o}_1$

Pitch: about $\hat{o}_2$

Yaw: about $\hat{o}_3$

* inertial velocity vector of the orbital frame 0

$${}^0W_{0/N} = \Omega \hat{o}_2$$

* For a circular orbit, the constant orbit rate magnitude is given through Kepler's equation.

$$\Omega^2 = \frac{GM_E}{R_c^3}$$

* B frame is principal coordinate frame of the rigid body

$${}^B W_{B/0} = \begin{bmatrix} -\sin\theta & 0 & 1 \\ \sin\phi \cos\theta & \cos\phi & 0 \\ \cos\phi \cos\theta & -\sin\phi & 0 \end{bmatrix} \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}$$

$$[B0] = \begin{bmatrix} \cos\psi & \cos\psi & -\sin\theta \\ \sin\psi \cos\theta - \phi \sin\psi & \sin\psi \sin\theta + \phi \cos\psi & \sin\phi \cos\theta \\ \cos\psi \sin\theta + \phi \sin\psi & \cos\psi \cos\theta - \phi \cos\psi & \cos\phi \cos\theta \end{bmatrix}$$

$${}^B W_{0/N} = [B0] {}^0 W_{0/N} = [B0] (\Omega \hat{o}_2) = \Omega \begin{pmatrix} \cos\psi \\ \sin\psi \sin\theta + \phi \cos\psi \\ \cos\psi \cos\theta - \phi \cos\psi \end{pmatrix}$$

$$\rightarrow {}^B W_{B/N} = {}^B W_{B/O} + {}^B W_{O/N}$$

$$= {}^B \begin{pmatrix} \dot{\phi} - s\theta\dot{\psi} + \Omega c\theta s\psi \\ s\phi c\theta\dot{\psi} + c\phi\dot{\theta} + \Omega(s\phi s\theta s\psi + c\phi c\psi) \\ c\phi c\theta\dot{\psi} - s\phi\dot{\theta} + \Omega(c\phi s\theta s\psi - s\phi c\psi) \end{pmatrix}$$

* Orbital frame에 대해 small rotations를 본다.

$$\textcircled{W_{B/N}} = W = {}^B \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \approx {}^B \begin{pmatrix} \dot{\phi} + \Omega\psi \\ \dot{\theta} + \Omega \\ \dot{\psi} + \Omega\phi \end{pmatrix}$$

↳ 결국 이 모든 라캅은 ${}^B W_{B/N}$ 을 구하기 위한 목적이다.

$$\textcircled{\dot{W}} = \frac{{}^B d}{dt}(W) + W \times W \approx {}^B \begin{pmatrix} \ddot{\phi} + \Omega\dot{\psi} \\ \ddot{\theta} \\ \ddot{\psi} + \Omega\dot{\phi} \end{pmatrix}$$

* W 와 $\dot{W}$ 를 구해서 나중에 $[I]\textcircled{\dot{W}} = -[\textcircled{\dot{W}}][I]W + \textcircled{L_c}$ 에 대입 가능
 $\downarrow$
 principal inertias

* external gravity gradient torque expression (L_G)

$$R_c = {}^B \begin{pmatrix} R_{c1} \\ R_{c2} \\ R_{c3} \end{pmatrix} = [B0] \begin{pmatrix} 0 \\ 0 \\ R_c \end{pmatrix} = {}^B \begin{pmatrix} -\sin\theta \\ \sin\phi \cos\theta \\ \cos\phi \cos\theta \end{pmatrix} R_c$$

이제 gravity gradient torque 식에 대입

$$L_G = \begin{pmatrix} L_{G1} \\ L_{G2} \\ L_{G3} \end{pmatrix} = \frac{3GM_e}{R_c^5} {}^B \begin{pmatrix} R_{c2} R_{c3} (I_{33} - I_{22}) \\ R_{c1} R_{c3} (I_{11} - I_{33}) \\ R_{c1} R_{c2} (I_{22} - I_{11}) \end{pmatrix}$$

$${}^B L_G = \frac{3}{2} \Omega^2 \begin{pmatrix} (I_{33} - I_{22}) \cos^2\theta \sin 2\phi \\ -(I_{11} - I_{33}) \cos\phi \sin 2\theta \\ -(I_{22} - I_{11}) \sin\phi \sin 2\theta \end{pmatrix} : \text{yaw angle } (\theta) \text{ 에 의존하지 않는다는 특징은 가짐}$$

$$\approx 3\Omega^2 \begin{pmatrix} (I_{33} - I_{22}) \phi \\ -(I_{11} - I_{33}) \theta \\ 0 \end{pmatrix}$$

* Equations of motion

$$[I]\dot{W} = -[\tilde{W}][I]W + L_c \text{ 에 } \dot{W}, W, L_c \text{ 대입}$$



$$I_{11}(\ddot{\phi} + \Omega\dot{\psi}) = -(I_{33} - I_{22})(\dot{\theta} + \Omega)(\dot{\psi} - \Omega\phi) + 3\Omega^2(I_{33} - I_{22})\phi$$

$$I_{22}\ddot{\theta} = -(I_{11} - I_{33})(\dot{\psi} - \Omega\phi)(\dot{\phi} + \Omega\psi) - 3\Omega^2(I_{11} - I_{33})\theta$$

$$I_{33}(\ddot{\psi} - \Omega\dot{\phi}) = -(I_{22} - I_{11})(\dot{\phi} + \Omega\psi)(\dot{\theta} + \Omega)$$

① pitch equation

$$\ddot{\theta} + 3\Omega^2 \left(\frac{I_{11} - I_{33}}{I_{22}} \right) \theta = 0 : \text{yaw와 roll 성분에 대해 decoupled 되었다}$$

방정식의 형태가 spring-mass system과 동일하므로 spring stiffness ≥ 0 이어야 복원력이 작용.

$$3\Omega^2 \left(\frac{I_{11} - I_{33}}{I_{22}} \right) \geq 0. \quad (I_{11} \geq I_{33})$$

② yaw and pitch equation.

$$k_R = \frac{I_{22} - I_{11}}{I_{33}}, \quad k_Y = \frac{I_{22} - I_{33}}{I_{11}} \quad \text{이러 하면}$$

$$\begin{pmatrix} \ddot{\phi} \\ \ddot{\psi} \end{pmatrix} + \begin{bmatrix} 0 & \Omega(1 - k_Y) \\ \Omega(k_R - 1) & 0 \end{bmatrix} \begin{pmatrix} \dot{\phi} \\ \dot{\psi} \end{pmatrix} + \begin{bmatrix} 4\Omega^2 k_Y & 0 \\ 0 & \Omega^2 k_R \end{bmatrix} \begin{pmatrix} \phi \\ \psi \end{pmatrix} = 0$$

↓

ϕ 와 ψ 가 coupled 되어 있으므로 특성방정식을 사용해야함.

(coupled linear time-invariant system)

↓

b_1

b_0

$$\lambda^4 + \lambda^2 \Omega^2 (1 + 3k_Y + k_Y k_R) + 4\Omega^4 k_Y k_R = 0$$

(여기서 양수인 실수 해가 발생하지 않을 때, 이 system이 stable)

* yaw-roll motion stability condition.

1) $k_R k_Y > 0$

$b_0 > 0$

2) $1 + 3k_Y + k_Y k_R > 0$

$b_1 > 0$

3) $(1 + 3k_Y + k_Y k_R)^2 > 16 k_Y k_R$

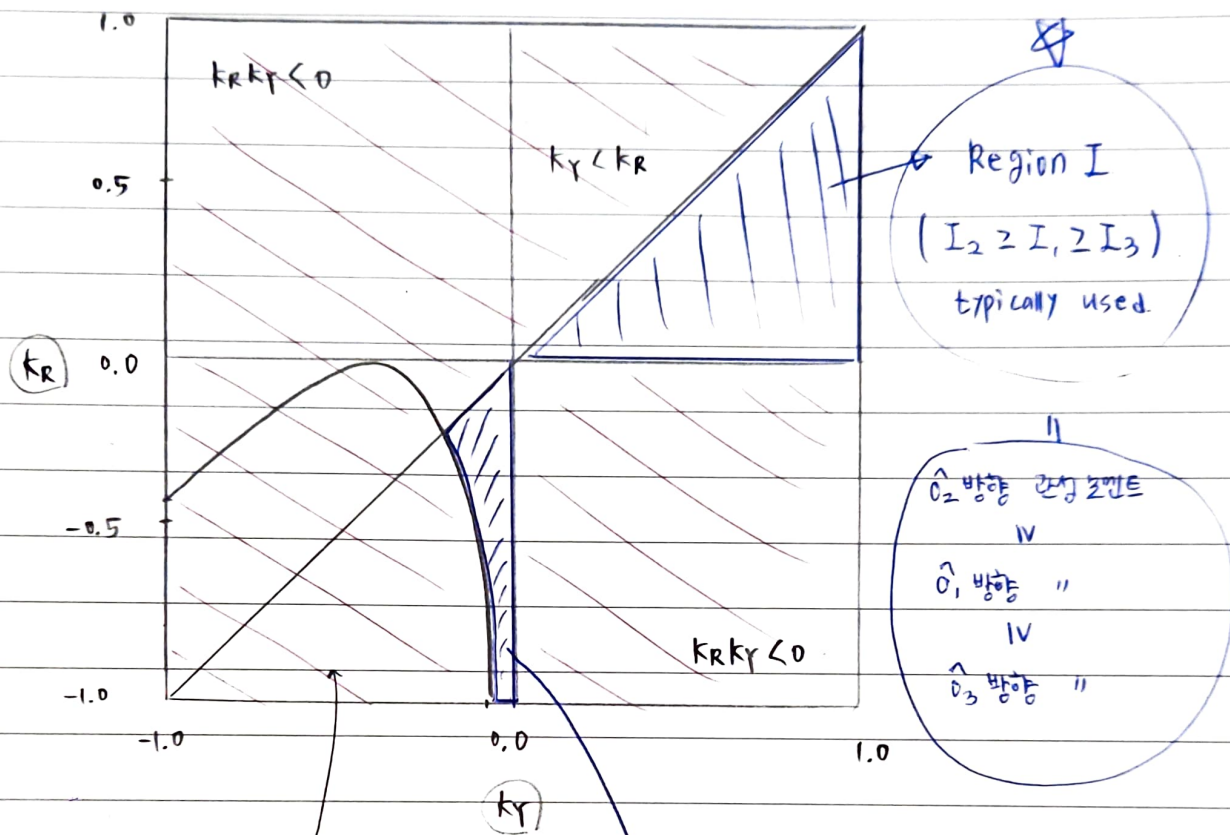
$\Leftrightarrow$

$b_1^2 - 4b_0 > 0$

4) $k_Y > k_R$

$I_{11} > I_{33}$

* Yaw, pitch, roll의 stability를 모두 고려한 그래프



* 파란색 부분이 안정한 부분. space shuttle의 경우 앞뒤로 길기 때문에 그만큼 pitch 방향 inertia가 크다.
(I_2)

* 여기서 inertia tensor 요소들이 body가 아닌 orbital frame 기준으로 정해져야 하는 이유는 I 의 값이 $\hat{O}_1, \hat{O}_2, \hat{O}_3$ 를 나타내기 때문. body 기준이라면 axes는 orbital과 동일하게 맞추기만 하면 되므로 I_1, I_2, I_3 사이에 아무런 상관관계가 없다. 따라서 $\begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$ 기준